Hybrid Clustering for Customer Segmentation Using RFM Analysis

Abstract

Customer segmentation is a critical task in modern data analytics, enabling businesses to understand customer behaviour and design targeted strategies. Traditional clustering algorithms such as K-Means, DBSCAN, and Fuzzy C-Means (FCM) are widely used for segmentation but suffer from individual limitations including sensitivity to noise, rigid cluster boundaries, and inability to capture overlapping patterns.

This paper proposes a hybrid clustering framework that integrates multiple clustering techniques to improve segmentation quality. Using an online retail dataset, customer behaviour is modelled through Recency, Frequency, and Monetary (RFM) features. The proposed approach includes combinations of K-Means, FCM, and DBSCAN using voting-based and noise-aware strategies.

The performance of individual and hybrid models is evaluated using a manually implemented silhouette score, demonstrating that hybrid clustering produces more cohesive, well-separated, and realistic customer segments compared to standalone algorithms.

1. Introduction

Customer segmentation is essential for understanding purchasing behaviour, improving customer retention, and increasing revenue. Clustering techniques are commonly used for this purpose as they group customers based on similarities without requiring labelled data.

However, no single clustering algorithm is universally optimal:

- K-Means is efficient but sensitive to noise and assumes spherical clusters
- DBSCAN detects outliers but struggles with varying density
- Fuzzy C-Means (FCM) handles overlapping clusters but lacks robustness

These limitations motivate the use of hybrid clustering, where multiple algorithms are combined to leverage their strengths and mitigate weaknesses.

2. Dataset Description

The dataset used in this study is an online retail dataset, containing transactional records of customers. Each row represents a purchase and includes:

- Customer ID
- Invoice number

- **Invoice date**
- **Quantity**
- **Unit price**

This dataset reflects real-world e-commerce behaviour, making it suitable for customer segmentation analysis.

3. Data Preprocessing

Data preprocessing ensures accuracy and reliability:

- **Removal of missing values**
- **Filtering invalid transactions (negative or zero values)**
- **Feature creation:**

$$\text{Total Amount} = \text{Quantity} \times \text{UnitPrice}$$

- **Conversion of invoice dates into datetime format**

4. RFM Feature Engineering

Customer behaviour is represented using the RFM model:

- **Recency (R): Days since last purchase**
- **Frequency (F): Number of transactions**
- **Monetary (M): Total spending**

This transforms transaction-level data into customer-level insights.

5. Data Normalization

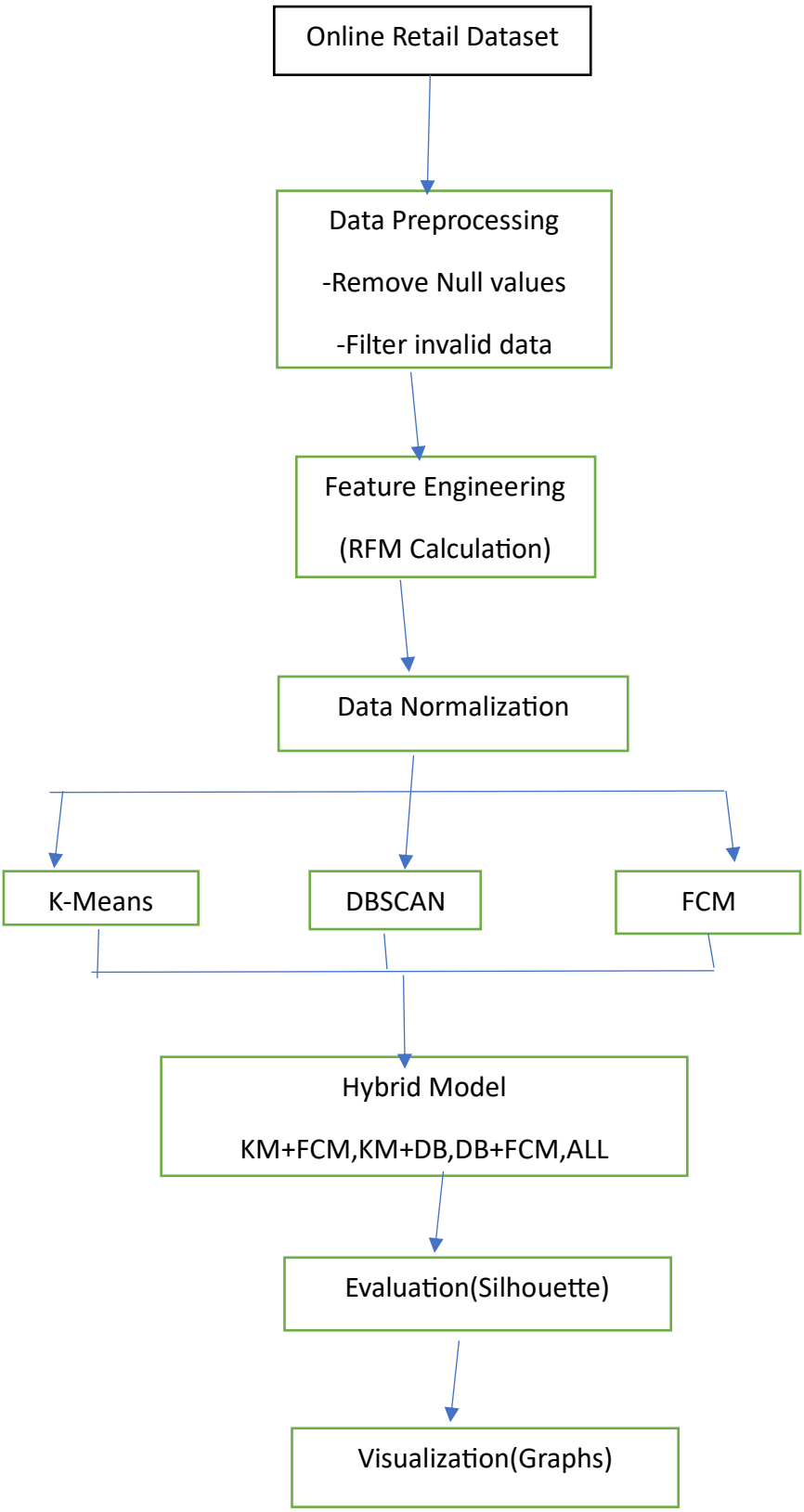
To ensure fair distance computation, features are standardized:

$$X = \frac{X - \mu}{\sigma}$$

This prevents any single feature (e.g., Monetary) from dominating clustering.

6.Architechture Diagram

The proposed system follows a pipeline-based architecture starting from data preprocessing to hybrid clustering and evaluation. Multiple clustering algorithms are applied in parallel, and their outputs are combined to generate robust customer segments.



7. Clustering Algorithms

7.1 K-Means

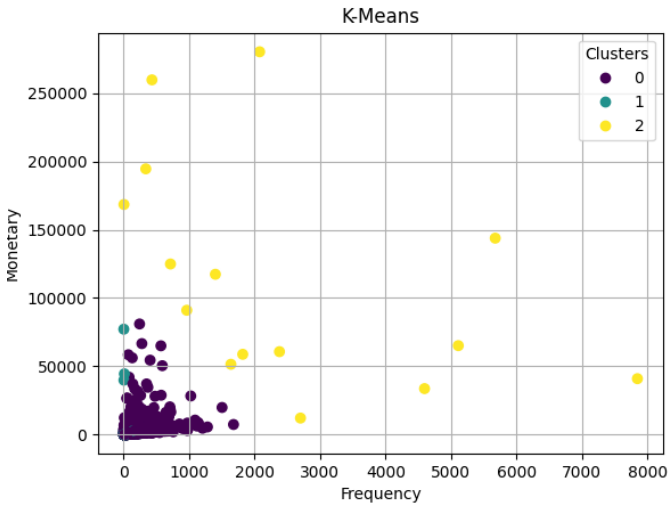
Partitions data into $k$clusters by minimizing intra-cluster variance.

Advantages:

- **Fast and efficient**

Limitations:

- **Sensitive to noise**
- **Requires predefined k**



7.2 DBSCAN

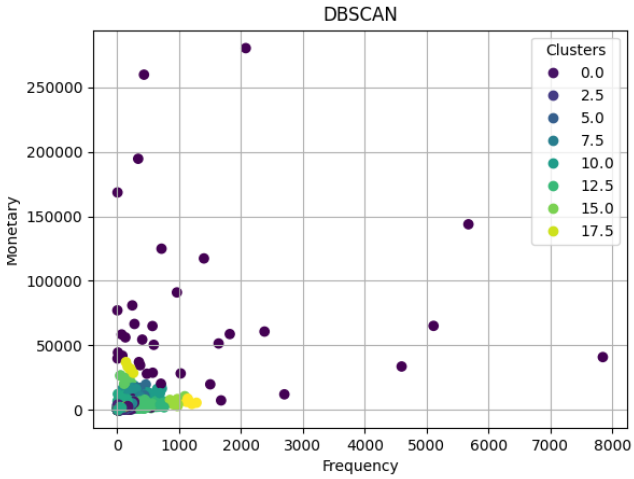
Clusters data based on density and identifies noise.

Advantages:

- **Detects outliers**
- **No need to specify number of clusters**

Limitations:

- **Sensitive to parameter selection**



7.3 Fuzzy C-Means (FCM)

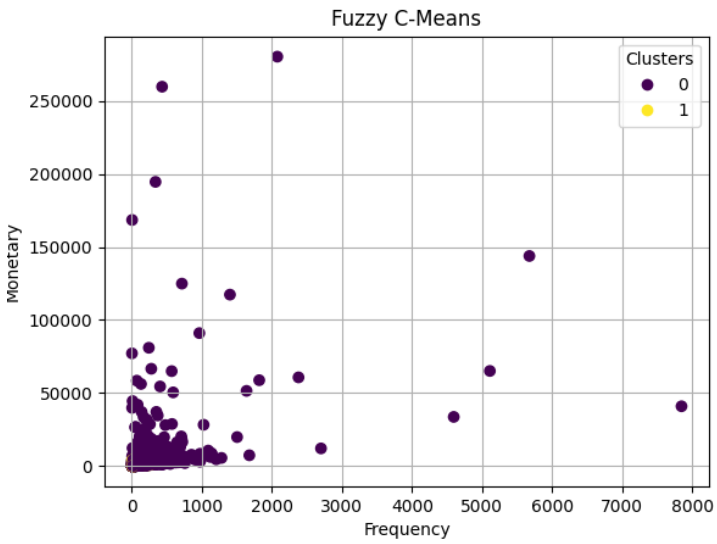
Assigns membership values to each data point across clusters.

Advantages:

- **Handles overlapping clusters**

Limitations:

- **Computationally expensive**
- **Sensitive to noise**



8. Proposed Hybrid Models

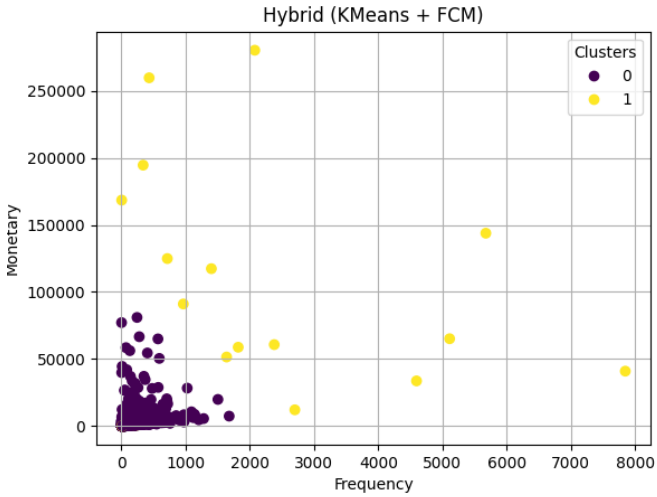
To overcome individual limitations, the following hybrid models are implemented:

8.1 K-Means + FCM

Combines hard and soft clustering using a voting mechanism.

Benefit:

- **Balances rigidity and flexibility**

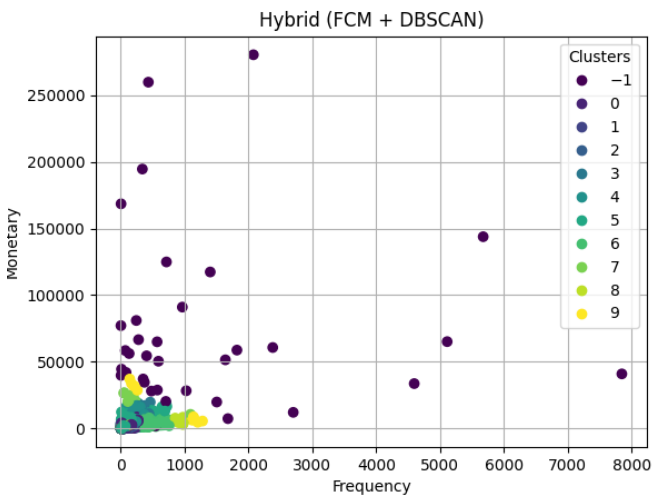


8.2 FCM + DBSCAN

Uses DBSCAN for noise detection and FCM for clustering.

Benefit:

- **Improved robustness by removing outliers**

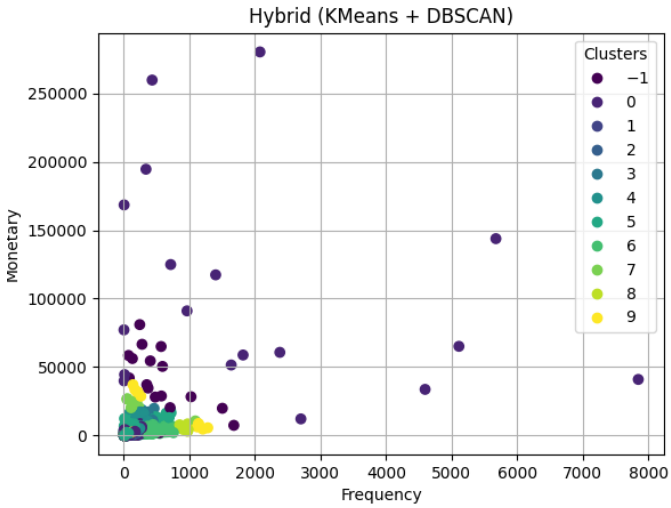


8.3 K-Means + DBSCAN

DBSCAN filters noise, and K-Means clusters the remaining data.

Benefit:

- **Cleaner and more stable clusters**



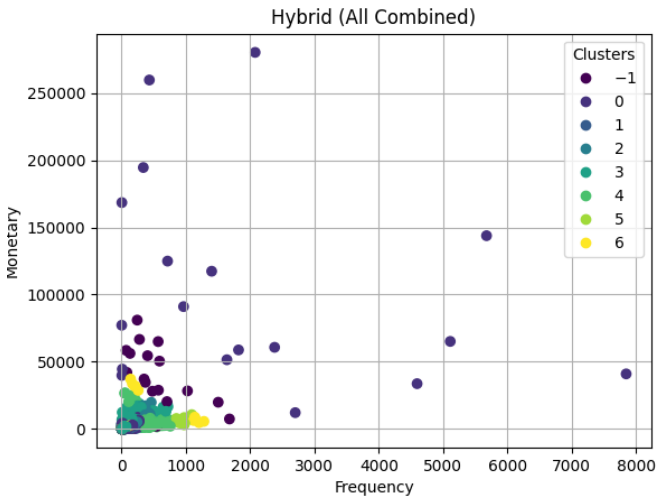
8.4 Combined Hybrid (K-Means + FCM + DBSCAN)

Integrates all three algorithms:

- **DBSCAN → noise handling**
- **K-Means → structured clustering**
- **FCM → flexibility**

Benefit:

- **Most comprehensive clustering approach**



9. Evaluation Method

Clustering performance is evaluated using the Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- $a(i)$**= intra-cluster distance**
- $b(i)$**= nearest-cluster distance**

A manual implementation is used to demonstrate conceptual understanding

10. Results and Analysis

The results indicate:

- **K-Means produces compact clusters but fails with noise**
- **DBSCAN effectively identifies outliers but may under-cluster**
- **FCM captures overlapping behaviour but is less stable**

Hybrid Model Performance

Hybrid models demonstrate:

- **Better separation between clusters**
- **Improved handling of noisy data**
- **More realistic customer grouping**

Silhouette scores show that hybrid approaches generally achieve higher values, indicating superior clustering quality.

11. Why Hybrid Clustering is Better

Hybrid clustering outperforms individual algorithms due to the following reasons:

11.1 Combines Strengths of Multiple Algorithms

- **K-Means → efficient structure**
- **FCM → flexible membership**
- **DBSCAN → noise detection**

Hybrid = best of all

11.2 Improved Accuracy

Individual models may misclassify points, whereas hybrid models reduce errors by combining multiple perspectives.

11.3 Effective Noise Handling

DBSCAN removes outliers that would otherwise distort clusters in K-Means and FCM.

11.4 Handles Overlapping Data

FCM allows partial membership, capturing real-world customer behaviour more accurately.

11.5 More Robust and Stable

Hybrid models are less sensitive to initialization and parameter variations.

12. Limitations

- **Increased computational complexity**
- **Requires careful design of combination strategy**
- **Dependent on parameter tuning**

13. Conclusion

This study demonstrates that hybrid clustering significantly improves customer segmentation compared to individual algorithms. By integrating K-Means, DBSCAN, and Fuzzy C-Means, the proposed framework achieves better clustering accuracy, robustness, and interpretability.

The use of RFM features effectively captures customer behaviour, and the manual implementation of silhouette score enhances conceptual understanding.

Hybrid clustering proves to be a powerful approach for real-world data analysis and customer analytics.

14. Future Work

- **Explore ensemble clustering techniques**
- **Automate parameter tuning**
- **Apply deep learning-based clustering**
- **Extend to real-time systems**